Data Structures and Algorithms

Beginner Study Notes and Practice Guide

This short guide introduces core concepts and provides practice ideas. It is an original learning summary, not a substitute for a complete textbook.

Introduction

Data structures organize information; algorithms describe steps for solving problems.

Arrays

Arrays store elements in indexed order. Index access is typically O(1), while searching an unsorted array is O(n).

Linked lists

Linked lists connect nodes using references. Insertions can be efficient when the relevant node is already known.

Stacks and queues

Stacks use last-in, first-out order. Queues use first-in, first-out order.

Searching

Linear search checks items one by one. Binary search is efficient on sorted data and takes O(log n) time.

Sorting

Sorting arranges values. Common algorithms include insertion sort, merge sort, and quicksort, with different trade-offs.

Complexity

Big O notation describes how resource use grows with input size.

Practice

Implement a stack, perform binary search on a sorted array, and compare linear and binary search.

Practice checklist

- Review the concepts and write a small example.
- Run your example and test an edge case.
- Explain the solution in your own words.